POSTDOCTORAL RESEARCHER IN GENERATIVE AI

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NICOLAS DUFOUR

PROFESSIONAL EXPERIENCE

UC Berkeley
Postdoctoral Researcher
2026–Present, Berkeley, CA, USA
Working with Alyosha Efros at Berkeley AI Research (BAIR).

Kyutai
Postdoctoral Researcher
January 2026–Present, Paris, France
Supervised by Patrick Pérez. Research on representation learning through generative models.

Meta
Applied Research Intern
July 2022-October 2022, London, UK
Supervised by Vincent MOENS.
Worked on integrating Model Based Reinforcement Learning to torchrl, a Pytorch domains library.

Ecole Nationale des Ponts et Chaussées (ENPC) and Ecole Polytechnique
Research Intern
April 2021-September 2021, Paris, France
Supervised by David PICARD and Vicky KALOGEITON.
Worked on subject transfer and conditional Generative Adversarial Networks (cGANs). Designed a novel semantically conditioned GAN architecture improving the state of the art. Joint internship between IMAGINE (ENPC) and GeoViC (Ecole Polytechnique) labs.

Upskills
Machine Learning Engineer Internship
February 2020-July 2020, Singapore
Worked on detecting fraudulent patterns between traders with anomaly detection for one of the largest Asian banks. Worked with Graph Neural Network in an unsupervised setting. Created a dataset from Murex trading data using Spark. Worked on novelty detection leveraging a Variational Auto-Encoder (VAE) architecture.

Boyce Thompson Institute - Cornell University
Research Internship
June 2018 / August 2018, Ithaca, NY, USA
Worked on a better way to store genotypic data. Tools such as Hadoop, Spark and the Parquet file format were used to optimize the storing performance. Developed some custom algorithms in Python to fit the data needs. Presented the results by a poster to a wide audience of researchers.

RESEARCH PAPERS

Rethinking Neural Thickets Without Weight Perturbations
arXiv preprint 2026

Nicolas Dufour, *Elouan Gardès, Amil Dravid, Patrick Pérez, Loic Landrieu*
Showed that adding a random bias to the output logits of a frozen model matches perturbing every weight.
Project page: https://kyutai.org/logit-psv/
Paper: https://arxiv.org/abs/2610.40649

Rethinking Video Autoencoders for Interactive World Models
arXiv preprint 2026

Adrien Ramanana Rahary, Yutong Bai, ***Nicolas Dufour****, Patrick Pérez, David Picard*
Introduced CADRE, open video autoencoders for interactive world models, and studied how much they should compress.
Project page: https://kyutai.org/blog/2026-10-22-cadre/
Paper: https://arxiv.org/abs/2610.30129

Generative Drifting is Secretly Score Matching: A Spectral and Variational Perspective
arXiv preprint 2026

Erkan Turan, ***Nicolas Dufour****, Maks Ovsjanikov*
Established the score-matching and Wasserstein-gradient-flow foundations of generative drifting and introduced bandwidth annealing for faster convergence.
Paper: https://arxiv.org/abs/2603.09936

RESEARCH PAPERS (CONTINUED)

Surflo: Consistent 3D Surface Flow Model with Global State

NeurIPS 2026 **Oral**

Antoine Guédon, Shu Nakamura*, **Nicolas Dufour***, Jiahui Lei, Ko Nishino, Angjoo Kanazawa*

Introduced a feed-forward model that encodes unposed views into one global state and reconstructs coherent 3D surfaces at arbitrary resolution.

Project page: https://anttwo.github.io/surflo/

Paper: https://arxiv.org/abs/2606.13644

One View Is Enough! Monocular Training for In-the-Wild Novel View Generation

NeurIPS 2026

*Adrien Ramanana Rahary, **Nicolas Dufour**, Patrick Pérez, David Picard*

Introduced OVIE, a novel-view generator trained on 30 million unpaired internet images using monocular geometry only as a training scaffold.

Project page: https://kyutai.org/blog/2026-04-14-ovie

Paper: https://arxiv.org/abs/2603.23488

Diffusable Latents from Structure-Agnostic Distillation

NeurIPS 2026 Workshop on Principles of Generative Modeling

*Adrien Ramanana Rahary, **Nicolas Dufour**, Patrick Pérez, David Picard*

Showed that distilling a single pooled image descriptor improves latent diffusability as well as dense distillation.

Project page: https://kyutai.org/blog/2026-09-28-structure-agnostic-distillation/

Paper: https://arxiv.org/abs/2609.39657

The FID Lottery: Quantifying Hidden Randomness in Generative-Model Evaluation

ECCV 2026 Workshop on Empirical Theory in Representation Learning (E.T.) **Oral**

***Nicolas Dufour**, Alexei A. Efros, Patrick Pérez*

Quantified the variability of FID across training and sampling seeds and proposed a more reliable multi-seed evaluation protocol.

Project page: https://kyutai.org/fid-lottery/

Paper: https://arxiv.org/abs/2606.20536

MIRO: MultI-Reward cOnditioned pretraining improves T2I quality and efficiency

ICML 2026

***Nicolas Dufour**, Lucas Degeorge*, Arijit Ghosh*, Vicky Kalogeiton†, David Picard†*

Proposed MIRO, conditioning T2I models on multiple reward models during training to improve visual quality and efficiency.

Project page: https://nicolas-dufour.github.io/miro/

Paper: https://arxiv.org/abs/2510.25897

PoM: A Linear-Time Replacement for Attention with the Polynomial Mixer

CVPR Findings 2026

*David Picard, **Nicolas Dufour**, Lucas Degeorge, Arijit Ghosh, Davide Allegro, Tom Ravaud, Yohann Perron, Corentin Sautier, Zeynep Sonat Baltaci, Fei Meng, Syrine Kalleli, Marta López-Rauhut, Thibaut Loiseau, Ségolène Albouy, Raphael Baena, Elliot Vincent, Loic Landrieu*

Introduced the Polynomial Mixer, a linear-complexity replacement for self-attention validated across text, vision, image generation, 3D modeling, and Earth observation.

Paper: https://arxiv.org/abs/2604.06129

DIPSY: Training-Free Synthetic Data Generation with Dual IP-Adapter Guidance

BMVC 2025

*Luc Boudier, Loris Manganelli, Eleftherios Tsonis, **Nicolas Dufour**, Vicky Kalogeiton*

Developed a training-free dual-guidance method for generating discriminative synthetic data from few-shot examples.

Paper: https://arxiv.org/abs/2509.22635

How far can we go with ImageNet for Text-to-Image generation?

arXiv preprint 2025

Lucas Degeorge, Arijit Ghosh*, **Nicolas Dufour**, David Picard†, Vicky Kalogeiton†*

We show that, contrary to the common belief, a small dataset of 1.2M images like ImageNet can be used to train a T2I model that matches or outperforms models trained on massive web-scraped collections.

Project page: https://lucasdegeorge.github.io/projects/t2i-imagenet/

Around the World in 80 Timesteps: A Generative Approach to Global Visual Geolocation

CVPR 2025

***Nicolas Dufour**, David Picard, Vicky Kalogeiton, Loic Landrieu*

Pioneered a probabilistic visual geolocation framework using diffusion and Riemannian flow matching that outperforms existing deterministic methods with improved accuracy and uncertainty quantification.

Project page: https://nicolas-dufour.github.io/plonk.html

E.T, the Exceptional Trajectories: Text-to-camera-trajectory generation with character awareness

ECCV 2024

*Robin Courant, **Nicolas Dufour**, Xi Wang, Marc Christie, Vicky Kalogeiton*

Proposed a novel dataset and diffusion-based approach to generate complex camera trajectories from textual captions, contributing to democratizing cinematography for common users.

Project page: https://www.lix.polytechnique.fr/vista/projects/2024_et_courant/

Analysis of Classifier-Free Guidance Weight Schedulers

TMLR 2024

*Xi Wang, **Nicolas Dufour**, Nefeli Andreou, Marie-Paule Cani, Victoria Fernandez Abrevaya, David Picard, Vicky Kalogeiton*

Analyzed classifier-free guidance weight schedulers and found that simple, monotonically increasing weights consistently improve performance and diversity of generated images with minimal code changes.

NICOLAS DUFOUR

RESEARCH PAPERS (CONTINUED)

Don't drop your samples! Coherence-aware training benefits Conditional diffusion

CVPR 2024 **Highlight (TOP 11%)**

Nicolas Dufour, *Victor Besnier, Vicky Kalogeiton, David Picard*

Proposed Coherence-Aware Diffusion to integrate coherence in conditional information, enabling models to learn from noisy annotations without discarding data.

Project page: https://nicolas-dufour.github.io/cad

OpenStreetView-5M: The Many Roads to Global Visual Geolocation

CVPR 2024

Guillaume Astruc, **Nicolas Dufour***, Ioannis Siglidis*, Constantin Aronssohn, Nacim Bouia, Stephanie Fu, Romain Loiseau, Van Nguyen Nguyen, Charles Raude, Elliot Vincent, Lintao XU, Hongyu Zhou, Loic Landrieu**

Introduced a large-scale, open-access dataset of 5.1 million geo-referenced street view images and benchmarked state-of-the-art image encoders, spatial representations, and training strategies.

Project page: https://osv5m.github.io/

SCAM! Transferring humans between images with Semantic Cross Attention Modulation

ECCV 2022

Nicolas Dufour, *David Picard, Vicky Kalogeiton*

Introduced SCAM for semantically conditioned image generation, focusing on subject transfer.

Project page: https://nicolas-dufour.github.io/scam

For older projects see https://github.com/nicolas-dufour

EDUCATION

Ecole Nationale des Ponts et Chaussées (ENPC) and Ecole Polytechnique

PhD

October 2021-November 2025, Paris, France

Supervised by David PICARD and Vicky KALOGEITON

Thesis: Controllability and Efficiency in Generative Models.

ENS Paris Saclay

MVA Master (Mathematics, Vision and Learning)

October 2020 to September 2021, Paris, France

Master of Science specialized in computer vision, applied mathematics and machine learning.

Télécom SudParis - Institut Polytechnique de Paris

Engineering diploma

September 2017 to September 2020, Evry, France

One of the top French Engineering Schools. Followed the MSA speciality (Modelisation, Statistics and Applications) which teaches the intricacies of statistical learning.

CPGE Joffre MPSI

MP speciality Computer Science

September 2014 to July 2017, Montpellier, France

Intensive preparation in Math and Physics for the highly competitive entrance exams to the French Grandes écoles. CPGE stands for Classes Préparatoires aux Grandes Écoles.

Lycée Français Paul Valéry de Cali

Baccalauréat with High Honors

September 2006 to July 2014, Cali, Colombia

Lived 8 years in Cali, Colombia and attend French School, a trilingual school (French, Spanish and English).

SKILLS

Languages spoken

- French (Bilingual)
- Spanish (Bilingual)
- English (990 points Toeic)
- Portuguese (Beginner)

Software proficiency

Google Cloud Computer, Photoshop, Inkscape, Premiere, Latex, SLURM.

Soft Skills

Project Management, Fast Learner, Communication, Problem solving, Entrepreneurial spirit, Curious, Creative

Programming language

Python (Pandas, Numpy, Scikit-learn, Matplotlib, Seaborn, Plotly, Hadoop, Spark, Plotly), Javascript (React, AngularJS, NodeJS, Socket.io), HTML5/CSS3, Bash, SQL, Caml.

Deep Learning Stack

Pytorch, Pytorch Lightning, Wandb, Hydra.

OPEN-SOURCE CONTRIBUTIONS

TorchRL
Worked on the model based aspect of the library

Torchmetrics
Improved the FID metric to work online, avoiding storing features.

Huggingface/diffusers
Implemented the sampling for K-LMS sampler

AWARDS & REVIEWS

Gold Reviewer Award — ICML 2026
Recognized for outstanding reviewing contribution

Outstanding Reviewer Award — CVPR 2026
Recognized for outstanding reviewing contribution

Outstanding Reviewer Award — NeurIPS 2025
Top 8% of reviewers

Outstanding Reviewer Award — ACCV 2022
Recognized for outstanding reviewing contribution

REVIEWS

Reviewer for ICML 2026, CVPR 2026, ICLR 2026, NeurIPS 2025, ICCV 2025, CVPR 2025, AAAI 2024, ECCV 2024, CVPR 2024, WACV 2024, ICCV 2023, and ACCV 2022.